

Ashok Choudhary

Mechanical Engineer | Tata Steel Maintenance & Automation | IIT Madras CFD Researcher | C++/Python/OpenFOAM

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Professional Summary

Mechanical Engineer combining **6 years of industrial maintenance, reliability, and automation experience at Tata Steel** with current **CFD and numerical simulation research at IIT Madras**. Skilled in plant equipment troubleshooting, RCA/FMEA, rotating machinery, ESPs, bag filters, conveyors, and automation retrofit projects, with hands-on experience in continuous-process steel-plant operations. Currently working on **OpenFOAM/PDRFoam solver modification in C++**, hydrogen-explosion safety modelling, Python/MATLAB post-processing, Linux, Git, Bash, and HPC-based simulation workflows. Also engaged part-time in **AI coding contribution and AI model evaluation at Outlier**, applying code review, debugging, STEM reasoning, and technical feedback skills. Seeking roles in **R&D, CFD/simulation, industrial digitalization, reliability engineering, and AI-assisted engineering**.

Core Skills for AI Coding & R&D Roles

Programming	Python, C++, MATLAB, Bash scripting, object-oriented programming, numerical programming, automation scripts
Code Review & Debugging	Bug isolation, error diagnosis, edge-case reasoning, reproducibility checks, correctness evaluation, technical feedback writing
AI / ML Tools	RAFT optical flow, SEA-RAFT optical flow, PyTorch workflow exposure, OpenCV, NumPy, pandas, Jupyter Notebook, Google Colab
STEM Reasoning	CFD, numerical methods, fluid mechanics, combustion modelling, turbulence modelling, experimental–computational validation
Developer Tools	Linux, Git/GitHub, VS Code, ParaView, HPC clusters, SLURM/PBS job scripts, file management, data pipelines
CAD & Design Tools	AutoCAD, CATIA, engineering drawing interpretation, mechanical component layout, equipment specification review
Communication	Technical writing, research presentation, guideline-based evaluation, remote/asynchronous work discipline

AI Coding & Model Evaluation Experience

AI Coding Contributor / AI Model Evaluation Outlier — Part-time / Remote	2026 – Present Remote
- Evaluate AI-generated coding and technical responses for correctness, clarity, reasoning quality, edge-case handling, and instruction following. - Apply Python/C++ knowledge, STEM reasoning, debugging skills, and engineering judgement to review model outputs and provide structured feedback. - Work on guideline-based evaluation tasks requiring clear written explanations, reproducible reasoning, and high attention to technical detail.	

Relevant Research Experience

CFD Researcher — Industrial Combustion & AI-assisted Flow Analysis Indian Institute of Technology Madras	Jun 2024 – Present Chennai, India
- Modified OpenFOAM's RANS-based PDRFoam solver in C++ to improve combustion model behaviour and support hydrogen explosion safety simulations. - Built Python and MATLAB post-processing pipelines for automated analysis, plotting, validation, and parametric comparison of CFD results. - Ran large-scale parallel simulations on national HPC systems such as PARAM Rudra / PARAM Shakti using Linux, shell scripting, and scheduler-based workflows. - Validated simulation predictions against multi-gate explosion experiments using systematic error analysis and experimental–computational comparison. - Applied RAFT/SEA-RAFT optical flow to extract velocity fields from combustion/flame videos and cross-check flow trends against CFD outputs. - Presented research at ICCMS 2025, explaining PDRFoam capabilities, CFD solver workflows, and physics-based simulation methods for industrial combustion and explosion-safety applications. - Presented research at ISHPMIE 2026; experienced in converting complex engineering results into clear written and visual explanations.	

Industrial Experience

Mechanical Maintenance Engineer
Tata Steel Limited — Integrated Steel Plant

Aug 2018 – Aug 2024
Jamshedpur, India

- Worked for **6 years** in a high-uptime industrial environment involving ESP units, bag filters, ID fans, conveyors, gearboxes, rotating machinery, and sinter plant auxiliaries.
- Conducted condition assessment, root-cause analysis, FMEA, failure diagnosis, and preventive/corrective maintenance of critical plant equipment.
- Led automation and upgradation projects including **IRRA technology deployment** and tray conveyor system improvement.
- Contributed to reliability improvements that supported production increase from **5,800 to 6,000 tonnes/day**; maintained strong safety and documentation discipline.

Selected Projects

Hydrogen Explosion CFD Validation using PDRFoam IIT Madras

- Simulated hydrogen-air explosion cases in congested geometries; compared CFD predictions with experimental benchmarks across 4-, 7-, and 9-gate configurations.

AI-based Optical Flow Velocity Extraction using RAFT/SEA-RAFT IIT Madras

- Used deep-learning optical flow methods on experimental and CFD video frames to extract velocity fields; developed Python workflows for frame processing, visualization, and CFD comparison.

OpenFOAM Solver Modification and HPC Workflow IIT Madras

- Edited solver-level C++ code, debugged compilation/runtime issues, managed OpenFOAM case files, and executed parallel simulations on multi-core HPC clusters.

IRRA Automation Retrofit & Conveyor Upgradation Tata Steel

- Integrated electromechanical retrofit technology and upgraded conveyor systems through component specification, troubleshooting, contractor coordination, and commissioning support.

Education

M.S. in Applied Mechanics (Research), IIT Madras	2027 Expected	8.59 CGPA
B.E. in Mechanical Engineering, Annamalai University	2012	9.18 CGPA

Publications, Presentations & Certifications

- **ISHPMIE 2026:** Paper accepted/presented at the International Symposium on Hazards, Prevention and Mitigation of Industrial Explosions.
- **ICCMS 2025:** Oral presentation at the International Conference on Computational & Mathematical Sciences.
- **Certifications:** AutoCAD; CATIA; AI and Career Empowerment — University of Maryland; YUVA AI for All — TCS iON; MATLAB & Simulink Workshop — IIT Madras; Google Colab Python — Great Learning.

Keywords

AI model training, AI coding contributor, code evaluation, debugging, LLM response evaluation, STEM reasoning, Python, C++, MATLAB, Bash, computational physics, CFD, OpenFOAM, PDRFoam, numerical simulation, Linux, Git, GitHub, HPC, SLURM, PBS, NumPy, pandas, OpenCV, PyTorch workflow, RAFT optical flow, SEA-RAFT optical flow, AutoCAD, CATIA, industrial maintenance, reliability engineering, automation retrofit, technical writing.